

Project 10 : Flammable Gas Detector

Introduction:

Pluto X2, can be transformed into a flying smoke alarm. MQ-2 and Buzzer are add-ons utilised for this project. We are going to see how to interface multiple add-ons on a Pluto drone.

Challenge:

Build a program for Pluto drone to make a flying Flammable gas leak alarm.

Add-On used:

The MQ-2 is a gas sensor used to detect gases such as Methane, Butane, LPG, Smoke, Alcohol.



Connection:

Connect the Buzzer_module on the RC port on Primus V5/X2 board
Connect the MQ-2 sensor on the ADC port of Primus V5/X2 board

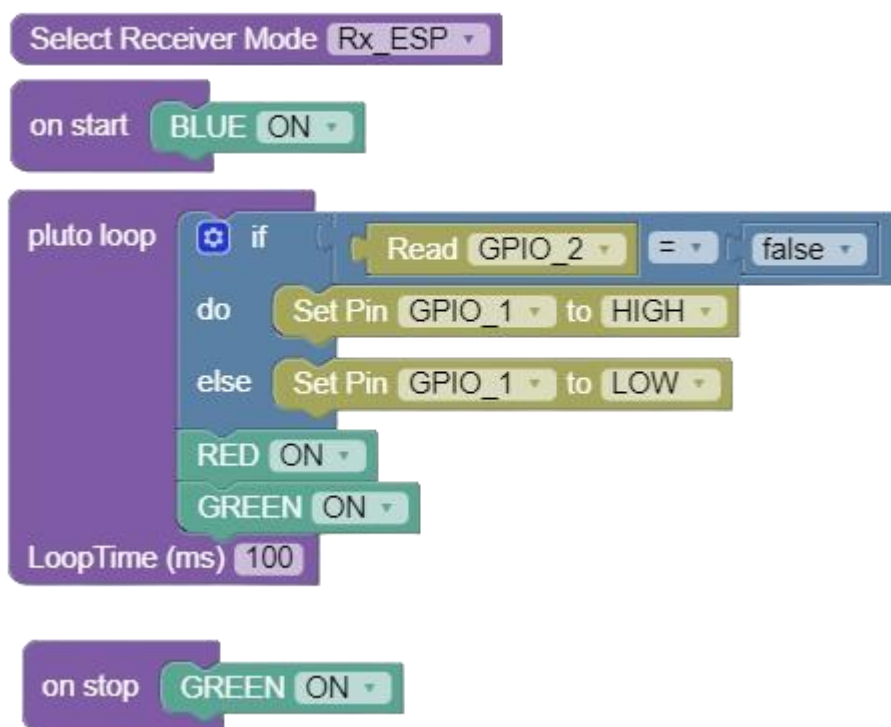
Primus V5/X2 pin	Peripherals
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VCC on ADC port	VCC on MQ-2
GND on ADC port	GND on MQ-2
Signal on ADC port	A0 on MQ-2
GND of RC port	GND of Buzzer
Signal of RC port	VCC of Buzzer

Approaching the Problem:

We utilise the MQ-2 sensor to detect flammable gas detector and to buzz an alarm.

Solution:



Explanation of Solution:

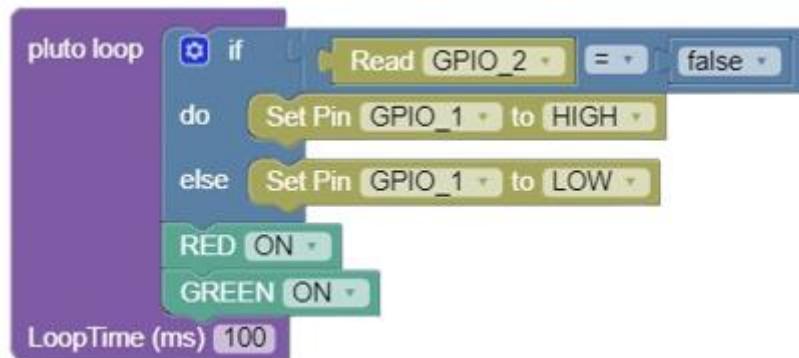
On Start Loop:

We set Blue Led HIGH



Pluto Loop:

In Pluto Loop we constantly keep checking, the GPIO_2 (ADC_port) on which MQ-2 is connected. MQ-2 goes LOW when it has detected smoke in its environment. So when it goes LOW, the GPIO_1 is made HIGH, that is Buzzer starts. In background Yellow light(Red+Green) acts as an indicator to Pluto Loop running.



On stop Loop: Green led turns on.





Fig : Pluto with MQ-2 Add-on

Try further!

- Make a fire alarm drone.
- Make a Gas leak alarm drone.